SOMALIA PREDICTORS

22/01/2024

Welcome to the data predictor documentation for Somalia. In this document, we provide a full description of the data sets used in the analysis. For a detailed overview of all available data sets and their respective sources, please refer to the Excel file "som_data_tracking.xlsx".

This document aims to provide an overview of the specific datasets used during the analysis study, giving readers a clear understanding of the data sources and their relevance to the research. For a more comprehensive view of the available datasets, including additional information and sources, we recommend consulting the accompanying Excel file.

To facilitate the analysis, we have provided an R script, "00_som_master_file.R," located in the "code" folder. Running this script will initiate the data cleaning, resulting in datasets essential for the analysis. But, before running the "00_som_master_file.R" script in the "code" folder, **it is crucial to ensure that the datasets are up-to-date**. After running the script, you will find the generated datasets saved in the "output" folder. Each dataset will include the following core elements: **year, p_time, month, date, region, district, pcode, reg_code**, and additional relevant information.

NB: *The number of district/admin2 in Somalia is 74.*
The beginning of the analysis is January 2015.

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Health Datasets

Malaria Dataset

Source: FSNAU Dashboard (directly from the website link, Fig.1)
Years: 2013 –> up-to-date
District: All
Malaria Incidence – Cases per 100,000 population

Measles Dataset

Source: FSNAU Dashboard (directly from the website, Fig.1)
Years: 2013 --> up-to-date
District: All
Measles Incidence – Cases per 100,000 population

Cholera Dataset

Source: FSNAU Dashboard (directly from the website, Fig.1)
Years: 2013 --> up-to-date
District: All
Cholera Incidence – Cases per 100,000 population

Humanitarian Presence

Insecurity Events

Source: https://acleddata.com/data-export-tool/ -- export the data + save it in the 'data' folder, Fig 2
Years: 2013 --> up-to-date
District: All
Saved as: acled_data_2023.csv
Incidence of insecurity events and fatalities – events of any typology per 100,000 population / fatalities per 100,000 population

Attacks Against Aid Workers

Source: https://aidworkersecurity.org – export the dataset and save it in the "data" folder, Fig.3
Years: 2013-2023
District: All
Saved as: security_incidents_2023.csv
Incidence of attacks against aid workers -- deaths/injuries per 100,000 population

Food Datasets

SAM/MAM

Source: https://drive.google.com/drive/folders/1Dl--uLDcgHhqhhe9qq5y0an9TelgKibe -- export the dataset and save it in the "data folder"
Years: 2013 --> up-to-date
District: All
Saved as: SOMALIA_NUTRITION_SERVICES_DATABASE_Sept_2023.xlsx
Malnutrition incidence -- cases of severe acute malnutrition admitted to treatment programmes per 100,000 population.

Trades/Power Index

Source: FSNAU Dashboard (directly from the website, Fig.1)
Years: 2013 --> up-to-date
District: All
Terms of trade purchasing power index -- Kcal equivalent of local cereals that an average local-quality goat (or daily labourer wage) can be exchanged for.

Water Price

Source: FSNAU Dashboard (directly from the website, Fig.1)
Years: 2013 --> up-to-date
District: All
Water price – price of 200L drum of water in Somali Shillings

Climate Datasets

Rainfall & CDI

Source: FSNAU Dashboard (directly from the website, Fig.1)
Years: 2013 --> up-to-date
District: All
Extent of drought conditions – Combined Drought Index
Rainfall – mean of Standard Precipitation Index

Datasets Description

Overview of Datasets

< Dec-2023 >

Type to search ..

Regions	Districts	Combined Drought Index (CDI)	Rainfall	Vegetation cover (NDVI)	Price of water	Sorghum prices	Maize prices	Red Rice prices	Local goat prices	Wage labor	TOT (goat to cereals)	TOT (wage to cereals)	Cost of Minimum Basket (CMB)	New Admissions (GAM)	Measles Cases	AWD/cholera cases	AWD/Cholera deaths	Malaria Cases	Insecurity - Incidents	Insecurity - Fatalities
Awdal	Borama	0.85	4.92	0.37	7,000	8,250	7,000	8,000	438,000	70,200	55	9	2,064,650	2,833				125		
Awdal	Baki	2.39	6.01	0.33										357				49		
Awdal	Lughaye	2	7.84	0.29		8,375	8,000	8,000	390,000		47			474				22		
Awdal	Zeylac	1.76	8.84	0.27	6,000	8,000		9,000	500,000	125,000	56	14	1,289,625	477				68		
Woqooyi Galbeed	Hargeysa	1.25	1.58	0.32	1,400	6,000	6,000	7,000	450,000	70,000	64	10	1,827,575	1,276				45	1	6
Woqooyi Galbeed	Berbera	1.66	2.27	0.36										221					1	0
Woqooyi Galbeed	Gebiley	0.92	2.07	0.4	13,000	9,000	9,000	9,000	510,000	29,250	57	3		718				74		
Togdheer	Burco	1.76	3.43	0.42	12,000	6,000	6,000	5,375	400,000	45,000	74	8	1,894,090	2,764				46	1	0
Togdheer	Buuhoodle	1.7	2.68	0.38										1,498	1	10		45	2	0

Figure 1 - FSNAU DashBoard (for the month of Dec. 2023) - https://dashboard.fsnau.org/dashboard/index

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	event_id_cnty	event_date	year	time_preci	disorder_t	event_typ	sub_event	actor1	assoc_act	inter1	actor2	assoc_act	inter2	interactior	civilian_ta	iso	region	country	admin1	admin2	admin3	location	latitud
2	SOM42688	01-Nov-23	2023	1	Demonstra	Protests	Peaceful p	Protesters	Labor Gro	6			0	60		706	Eastern Af	Somalia	Galgaduud	Dhuusamarreeb		Dhuusama	5.5
3	SOM42718	01-Nov-23	2023	1	Political vi	Battles	Armed cla	Al Shabaab		2	Police Forces of Som		1	12		706	Eastern Af	Somalia	Banadir	Banadir		Mogadishu	2.0
4	SOM42719	01-Nov-23	2023	1	Political vi	Battles	Armed cla	Majeerteen-Omar Ma		4	Majeerteen-Omar Ma		4	44		706	Eastern Af	Somalia	Mudug	Jariiban		Jariiban	7.2
5	SOM42722	01-Nov-23	2023	1	Strategic d	Strategic d	Disrupted	Police Forces of Som		1	Al Shabaab		2	12		706	Eastern Af	Somalia	Banadir	Banadir		Mogadishu	2.0
6	SOM42724	01-Nov-23	2023	1	Political vi	Battles	Armed cla	Military Forces of Sor		1	Al Shabaab		2	12		706	Eastern Af	Somalia	Bakool	Xudur		Abaq Beed	4.1
7	SOM42725	01-Nov-23	2023	1	Political vi	Battles	Armed cla	Military Forces of Sor		1	Al Shabaab		2	12		706	Eastern Af	Somalia	Bakool	Xudur		Ceel Garas	4.1
8	SOM42726	01-Nov-23	2023	1	Political vi	Battles	Armed cla	Military Forces of Sor		1	Al Shabaab		2	12		706	Eastern Af	Somalia	Bakool	Xudur		Dudumaal	4.1
9	SOM42727	01-Nov-23	2023	1	Political vi	Battles	Armed cla	Military Forces of Sor		1	Al Shabaab		2	12		706	Eastern Af	Somalia	Bakool	Xudur		Hudur	4.1
10	SOM42713	31-Oct-23	2023	1	Political vi	Violence a	Attack	Habar Gedir-Ayr Sub-(		4	Civilians (S Habar Ged		7	47	Civilian tar	706	Eastern Af	Somalia	Galgaduud	Dhuusamarreeb		Dhuusama	5.5
11	SOM42715	31-Oct-23	2023	1	Political vi	Violence a	Attack	Habar Gedir-Saleban		4	Civilians (S Habar Ged		7	47	Civilian tar	706	Eastern Af	Somalia	Hiraan	Belet Weyne		Matabaan	5.1
12	SOM42716	31-Oct-23	2023	1	Political vi	Battles	Armed cla	Al Shabaab		2	ATMIS: African Union		8	28		706	Eastern Af	Somalia	Bakool	Waajid		Waajid	3.8
13	SOM42717	31-Oct-23	2023	1	Political vi	Battles	Armed cla	Al Shabaab		2	Military Forces of Sor		1	12		706	Eastern Af	Somalia	Lower Sha	Marka		Dhanaane	1.8
14	SOM42720	31-Oct-23	2023	1	Political vi	Battles	Armed cla	Military Forces of Sor		1	Al Shabaab		2	12		706	Eastern Af	Somalia	Lower Jub	Kismaayo		Beerhani	-0.2
15	SOM42721	31-Oct-23	2023	1	Political vi	Battles	Armed cla	Military Forces of Sor		1	Al Shabaab		2	12		706	Eastern Af	Somalia	Lower Jub	Kismaayo		Kismayo	-0.3
16	SOM42689	30-Oct-23	2023	1	Political vi	Battles	Armed cla	Military Fc	Military Fc	1	Al Shabaab		2	12		706	Eastern Af	Somalia	Lower Jub	Kismaayo		Buulo Xaaj	-0.
17	SOM42701	30-Oct-23	2023	1	Political vi	Battles	Armed cla	Habar Gedir-Ayr Sub-(		4	Habar Gedir-Saleban		4	44		706	Eastern Af	Somalia	Galgaduud	Cadaado		Xananburr	5
18	SOM42705	30-Oct-23	2023	1	Political vi	Explosions	Remote ex	Unidentified Armed G		3	Civilians (S Labor Gro		7	37	Civilian tar	706	Eastern Af	Somalia	Banadir	Banadir		Mogadishu	2.0
19	SOM42707	30-Oct-23	2023	1	Political vi	Battles	Armed cla	Al Shabaab		2	Military Forces of Sor		1	12		706	Eastern Af	Somalia	Banadir	Banadir		Mogadishu	2.0
20	SOM42708	30-Oct-23	2023	1	Political vi	Battles	Armed cla	Al Shabaab		2	Military Forces of Sor		1	12		706	Eastern Af	Somalia	Banadir	Banadir		Mogadishu	2.
21	SOM42709	30-Oct-23	2023	1	Strategic d	Strategic d	Change to	Al Shabaab		2			0	20		706	Eastern Af	Somalia	Hiraan	Belet Weyne		Belet Wey	4.

Figure 2- View of ACLED Datasets – Important columns: event_date, event_type, admin2 -- https://acleddata.com/data-export-tool/

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Incident ID	Year	Month	Day	Country Co	Country	Region	District	City	UN	INGO	ICRC	NRCS and	NNGO	Other	Nationals	Nationals	Nationals	Total natic	Internatio	Internatio	Internatio	Total inter
	#event+id	#date+yea	#date+mo	#date+day	#country+	#country+	#adm1+na	#adm2+na	#adm3+na	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+	#affected+
	35	1997								0	0	2	0	0	0	1	0	0	1	0	1	0	1
	1	1997	1		KH	Cambodia	Banteay	Meanchey		0	0	1	0	0	0	1	0	0	1	0	0	0	0
	2	1997	1		RW	Rwanda	Northern	Musanze	Ruhengeri	0	4	0	0	0	0	0	0	0	0	3	1	0	4
	3	1997	2		TJ	Tajikistan				4	0	2	0	0	0	0	0	4	4	0	0	2	2
	4	1997	2		SO	Somalia	Lower Jub	Kismayo	Kismayo	0	1	0	0	0	0	0	0	0	0	1	0	0	1
	5	1997	2	14	RW	Rwanda	Kigali	Kigali	Kigali	1	0	0	0	0	0	1	0	0	1	0	0	0	0
	7	1997	5		CD	DR Congo				0	0	0	10	0	0	10	0	0	10	0	0	0	0
	6	1997	5	7	SL	Sierra Leone				3	0	0	0	0	0	1	2	0	3	0	0	0	0
	11	1997	6		SO	Somalia	Gedo	Baardheer	Tulo-Burag	1	0	0	0	0	0	1	0	0	1	0	0	0	0
	12	1997	6	8	RW	Rwanda	Northern	Musanze	Ruhengeri	0	1	0	0	0	0	1	0	0	1	0	0	0	0
	8	1997	6	14	RW	Rwanda	Northern	Musanze	Ruhengeri	1	0	0	0	0	0	1	0	0	1	0	0	0	0
	9	1997	6	17	RW	Rwanda	Northern	Musanze	Ruhengeri	1	0	0	0	0	0	1	0	0	1	0	0	0	0
	10	1997	6	19	RW	Rwanda	Northern	Musanze	Ruhengeri,	1	0	0	0	0	0	1	0	0	1	0	0	0	0
	14	1997	6	19	SO	Somalia	Lower Jub	Kismayo		0	1	0	0	0	0	1	0	0	1	0	0	0	0
	13	1997	6	20	SO	Somalia	Bay	Baydhaba		0	1	0	0	0	0	0	0	0	0	1	0	0	1
	16	1997	7			Chechnya				0	1	0	0	0	0	1	0	0	1	0	0	0	0
	17	1997	7			Chechnya				0	2	0	0	0	0	0	0	0	0	0	0	2	2

Figure 3 - View of the Aid workers datasets – Important columns: Incident ID, Year, Month, Country, District, Total killed, Total Wounded -- https://aidworkersecurity.org

	A	U	V	W	X
1	Data source	site_dwellers3	Partner current source of funds: cash and supplies	Add other source of funds:	site_activity/OTP
2	NC Somalia	n/a	n/a	n/a	TRUE
3	NC Somalia	n/a	n/a	n/a	TRUE
4	NC Somalia	n/a	n/a	n/a	TRUE
5	NC Somalia	n/a	n/a	n/a	TRUE
6	NC Somalia	n/a	n/a	n/a	TRUE
7	NC Somalia	n/a	n/a	n/a	FALSE
8	NC Somalia	n/a	n/a	n/a	TRUE
9	NC Somalia	n/a	n/a	n/a	TRUE
10	NC Somalia	n/a	n/a	n/a	TRUE
11	NC Somalia	n/a	n/a	n/a	TRUE
12	NC Somalia	n/a	n/a	n/a	TRUE
13	NC Somalia	n/a	n/a	n/a	TRUE
14	NC Somalia	n/a	n/a	n/a	TRUE
15	NC Somalia	n/a	n/a	n/a	TRUE
16	NC Somalia	n/a	n/a	n/a	FALSE
17	NC Somalia	n/a	n/a	n/a	TRUE
18	NC Somalia	n/a	n/a	n/a	FALSE
19	NC Somalia	n/a	n/a	n/a	TRUE
20	NC Somalia	n/a	n/a	n/a	TRUE
21	NC Somalia	n/a	n/a	n/a	TRUE
22	NC Somalia	n/a	n/a	n/a	TRUE
23	NC Somalia	n/a	n/a	n/a	TRUE
24	NC Somalia	n/a	n/a	n/a	TRUE
25	NC Somalia	n/a	n/a	n/a	TRUE
26	NC Somalia	n/a	n/a	n/a	TRUE
27	NC Somalia	n/a	n/a	n/a	FALSE
28	NC Somalia	n/a	n/a	n/a	TRUE
29	NC Somalia	n/a	n/a	n/a	TRUE
30	NC Somalia	n/a	n/a	n/a	TRUE
31	NC Somalia	n/a	n/a	n/a	TRUE
32	NC Somalia	n/a	n/a	n/a	TRUE
33	NC Somalia	n/a	n/a	n/a	TRUE
34	NC Somalia	n/a	n/a	n/a	TRUE

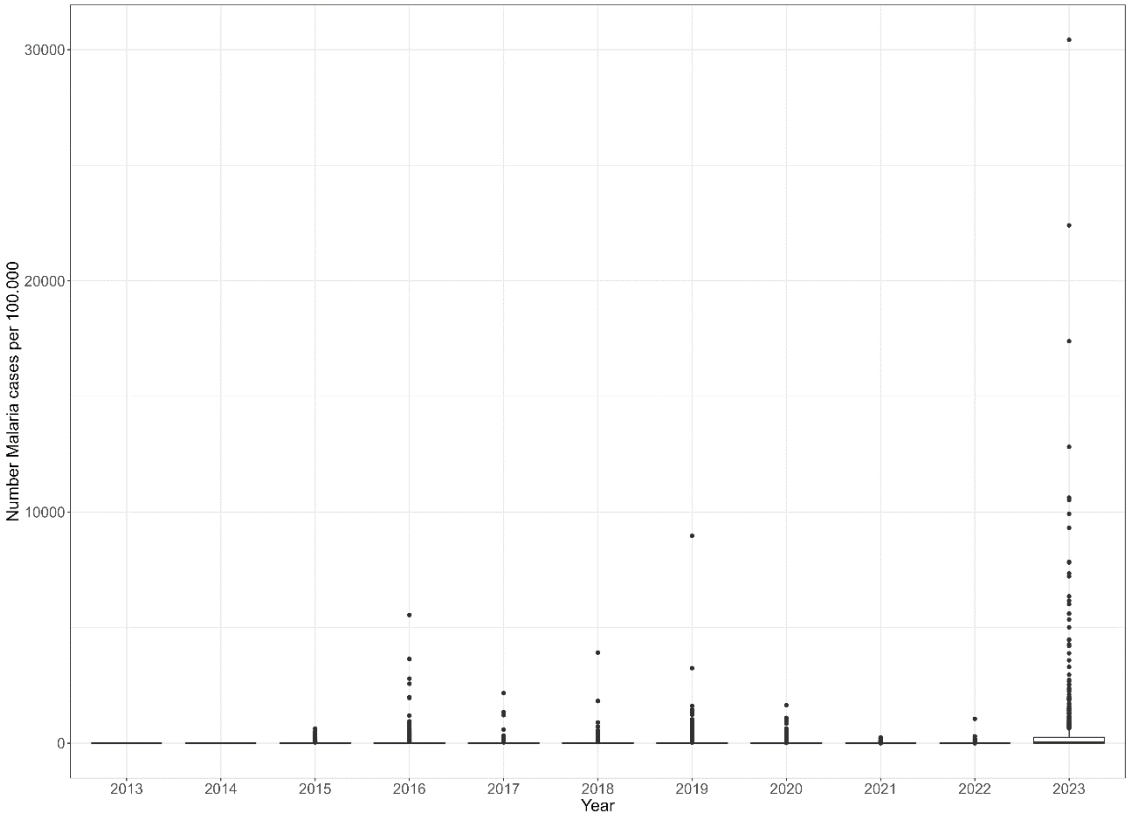
District|Partners | Admission Trend | DATA from ONA | LISTS | +

Figure 4 - View of the SAM Dataset -- Important sheet: DATA FROM ONA – Important columns: District, Year, Month, SAM#

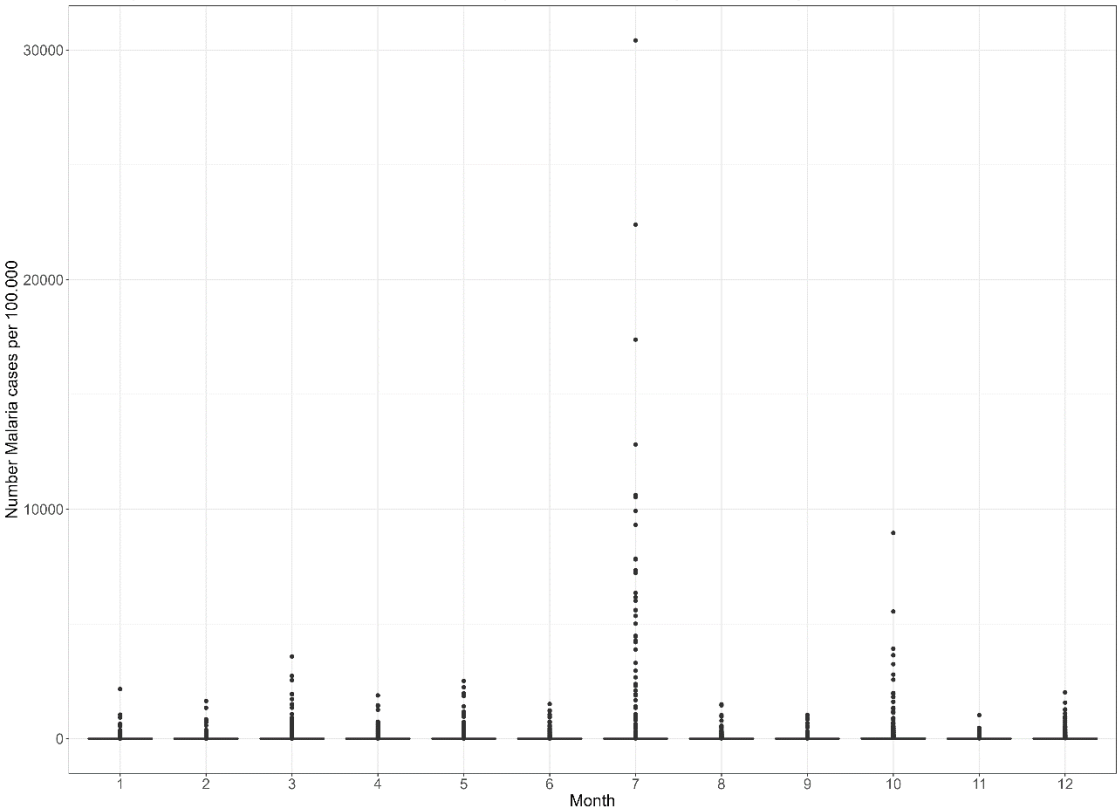
Dataset Visualizations

Malaria Cases

BBox plot of the number of malaria cases per year. Each point corresponds to a district.

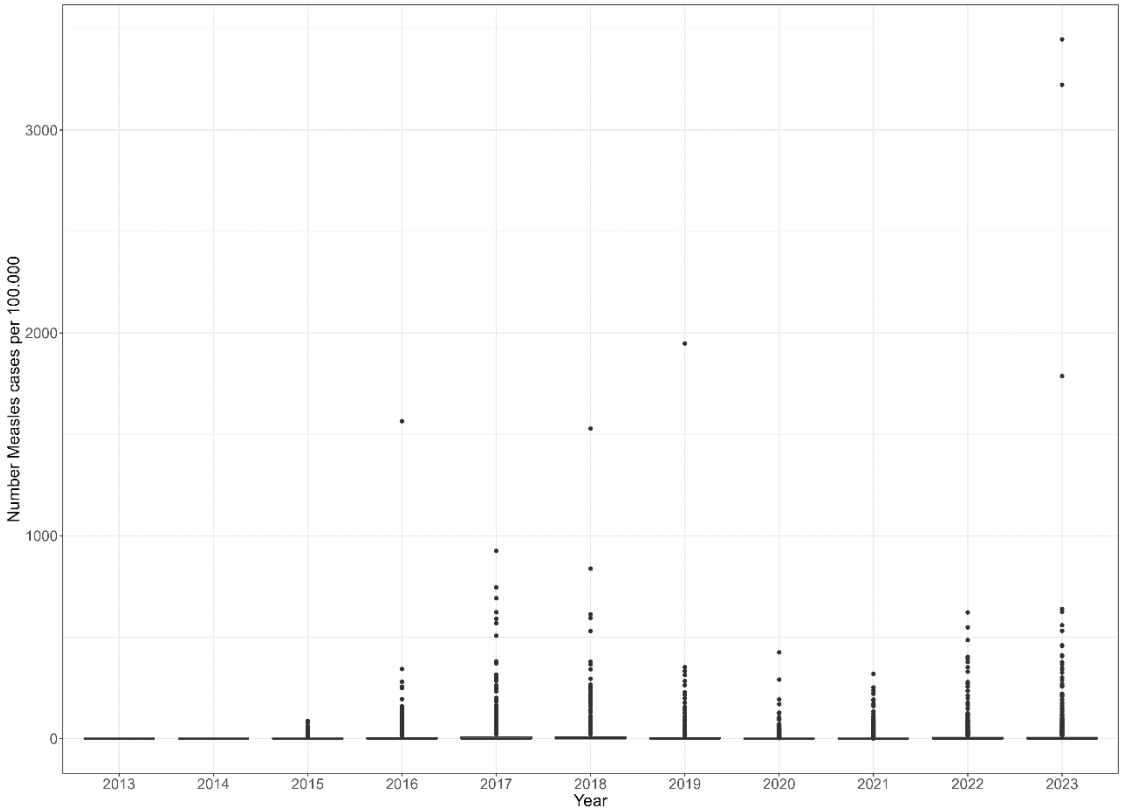


BBox plot of the number of malaria cases per month Each point corresponds to a district.

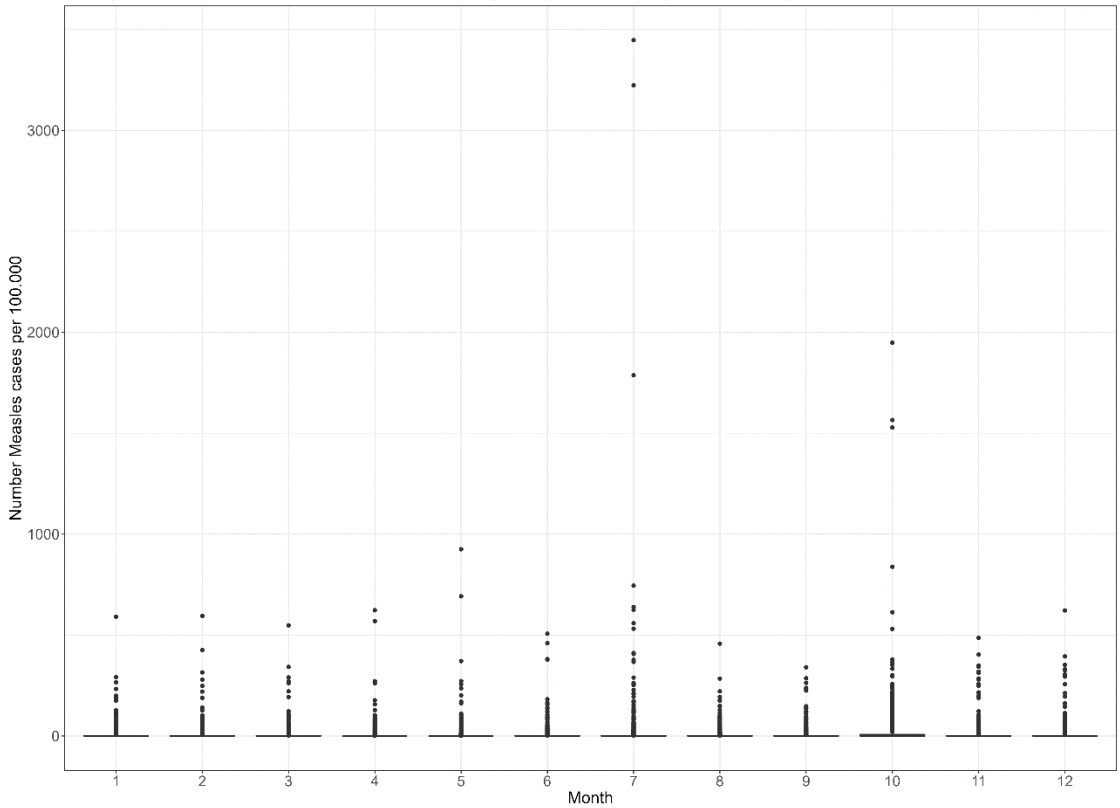


Measles Cases

Box plot of the number of measles cases per year. Each point corresponds to a district.

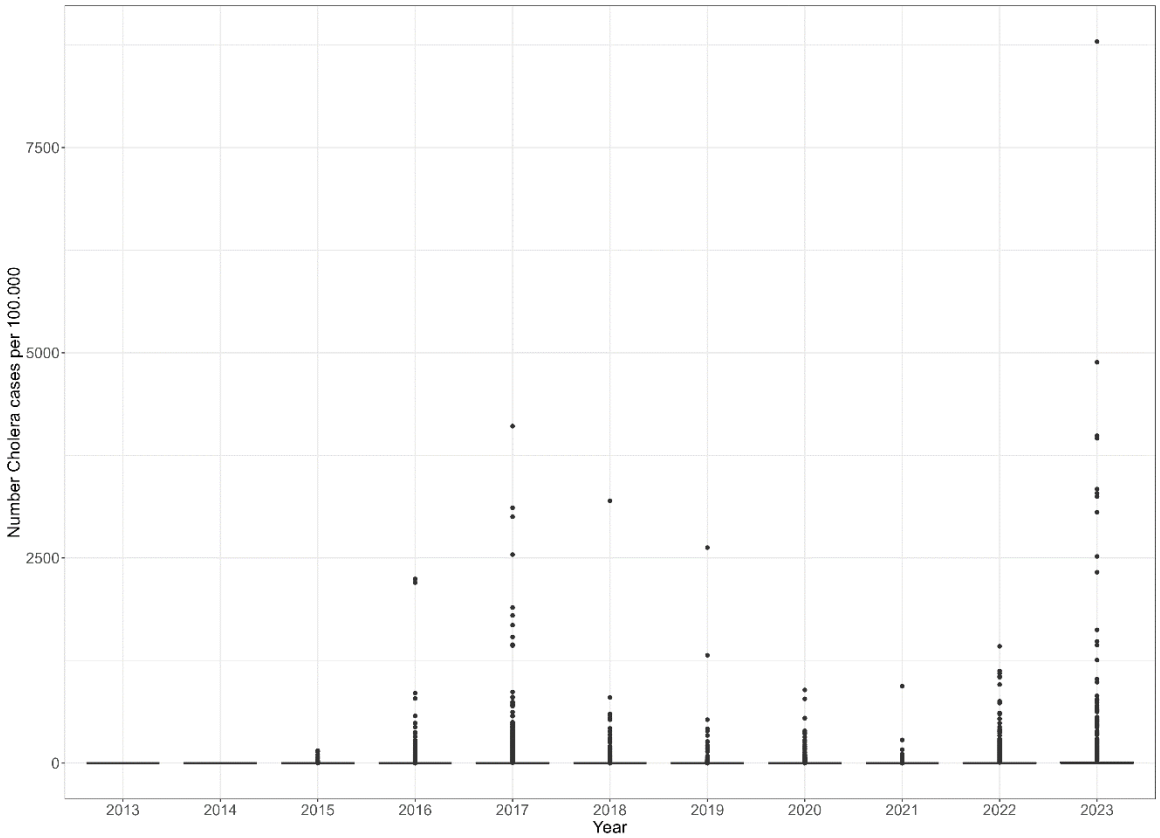


Box plot of the number of measles cases per month Each point corresponds to a district.

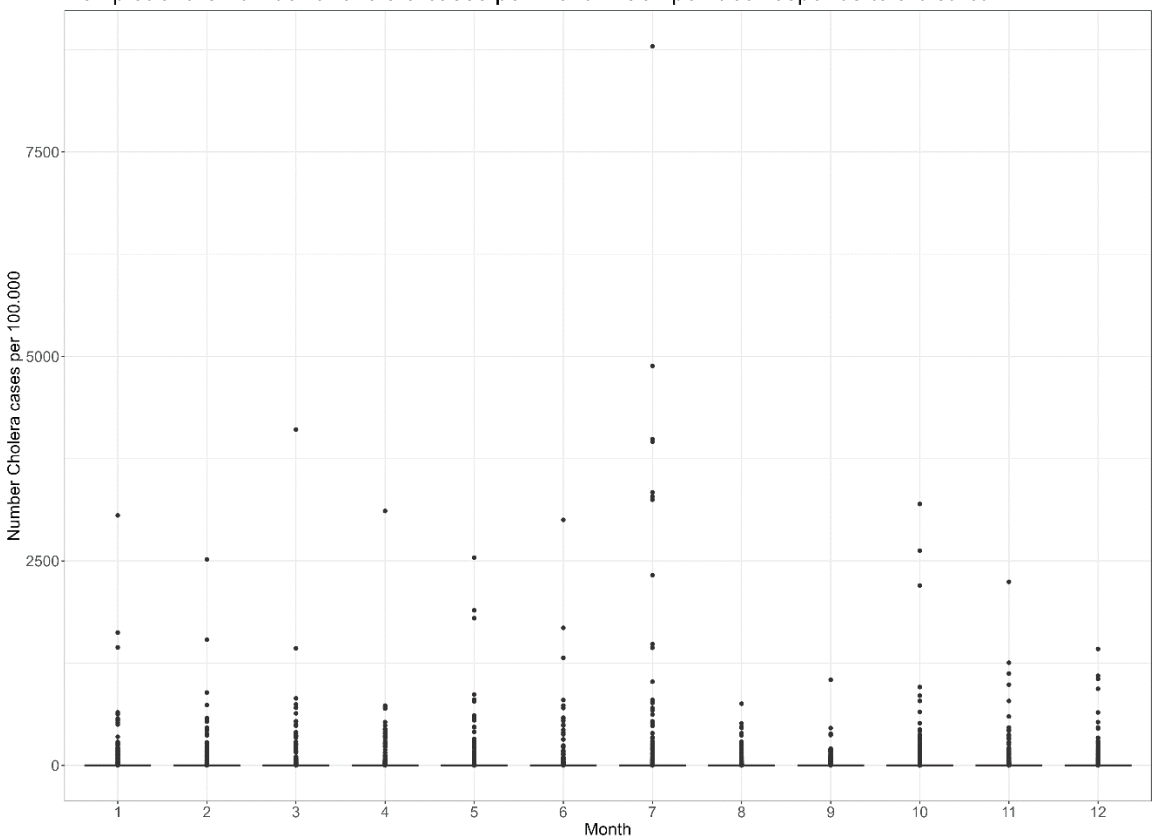


Cholera Cases

Box plot of the number of cholera cases per year. Each point corresponds to a district.

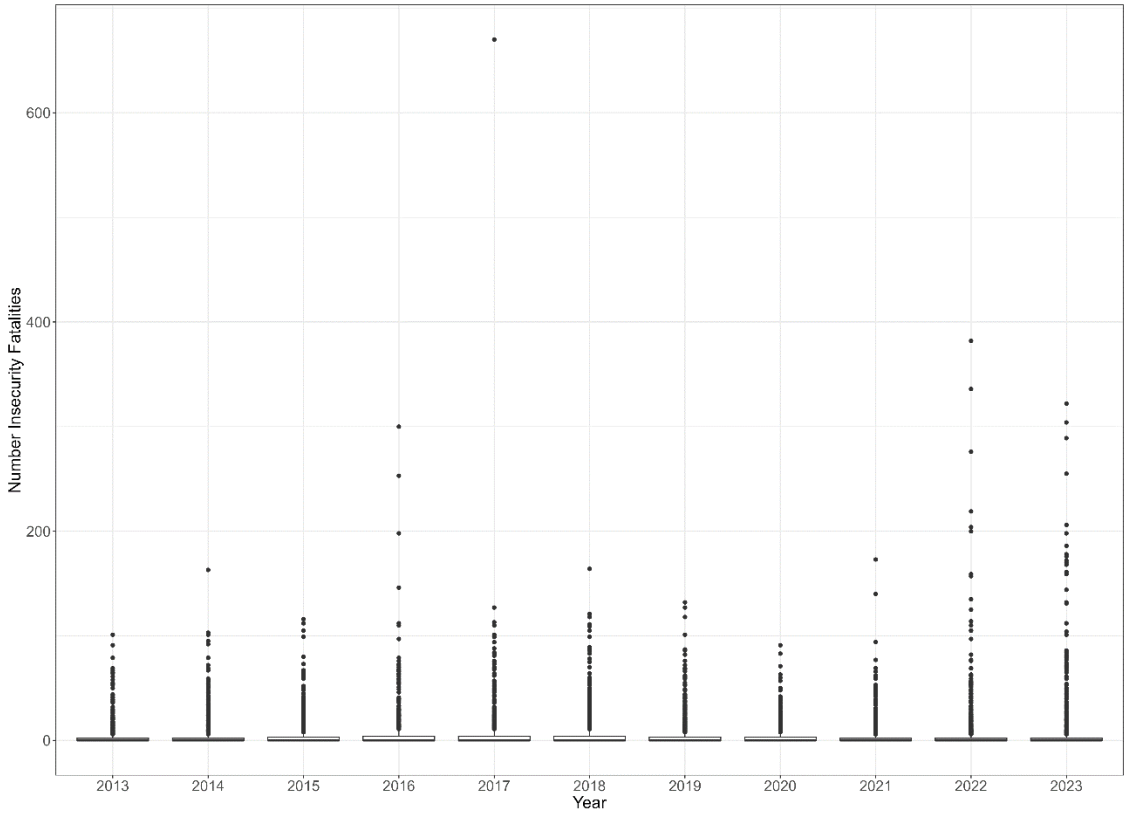


Box plot of the number of cholera cases per month Each point corresponds to a district.

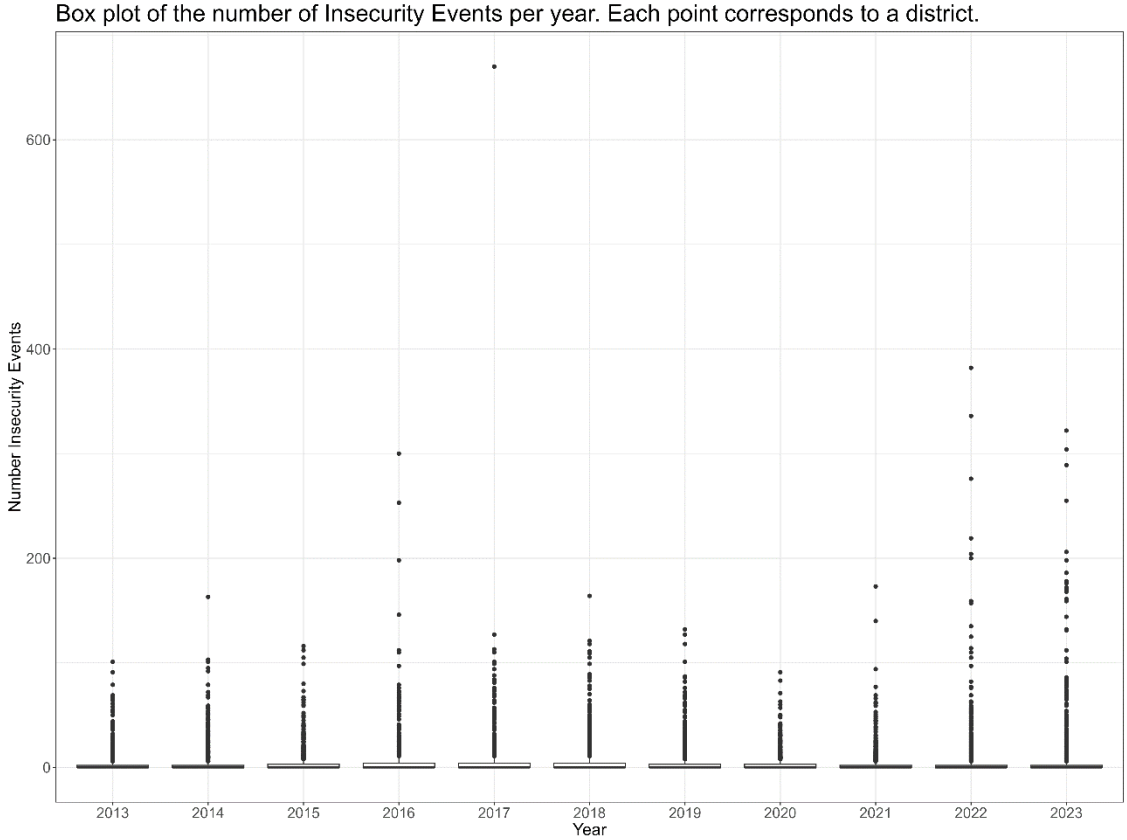


Insecurity Events

Box plot of the number of Insecurity Fatalities per year. Each point corresponds to a district.

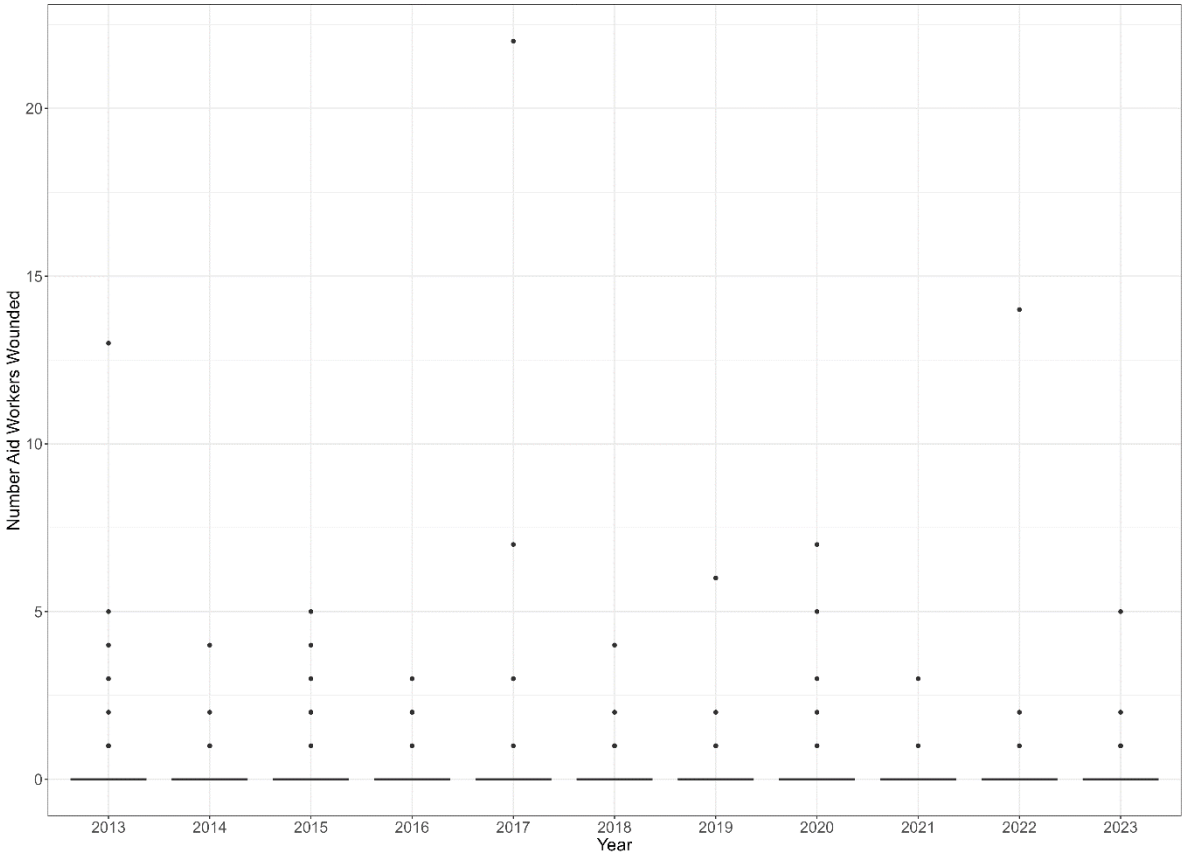


Box plot of the number of Insecurity Events per year. Each point corresponds to a district.

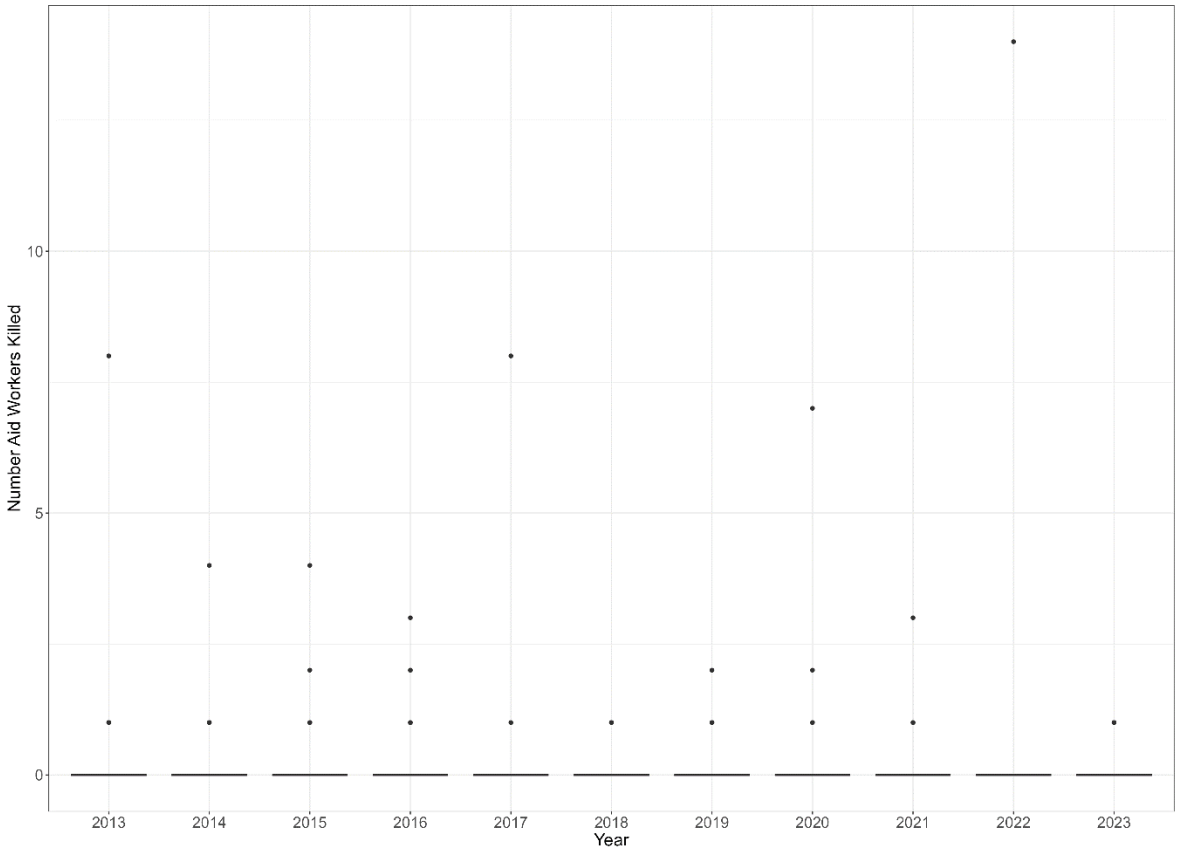


Attacks Against Aid Workers

Box plot of the number of Aid Workers Wounded per year. Each point corresponds to a district.

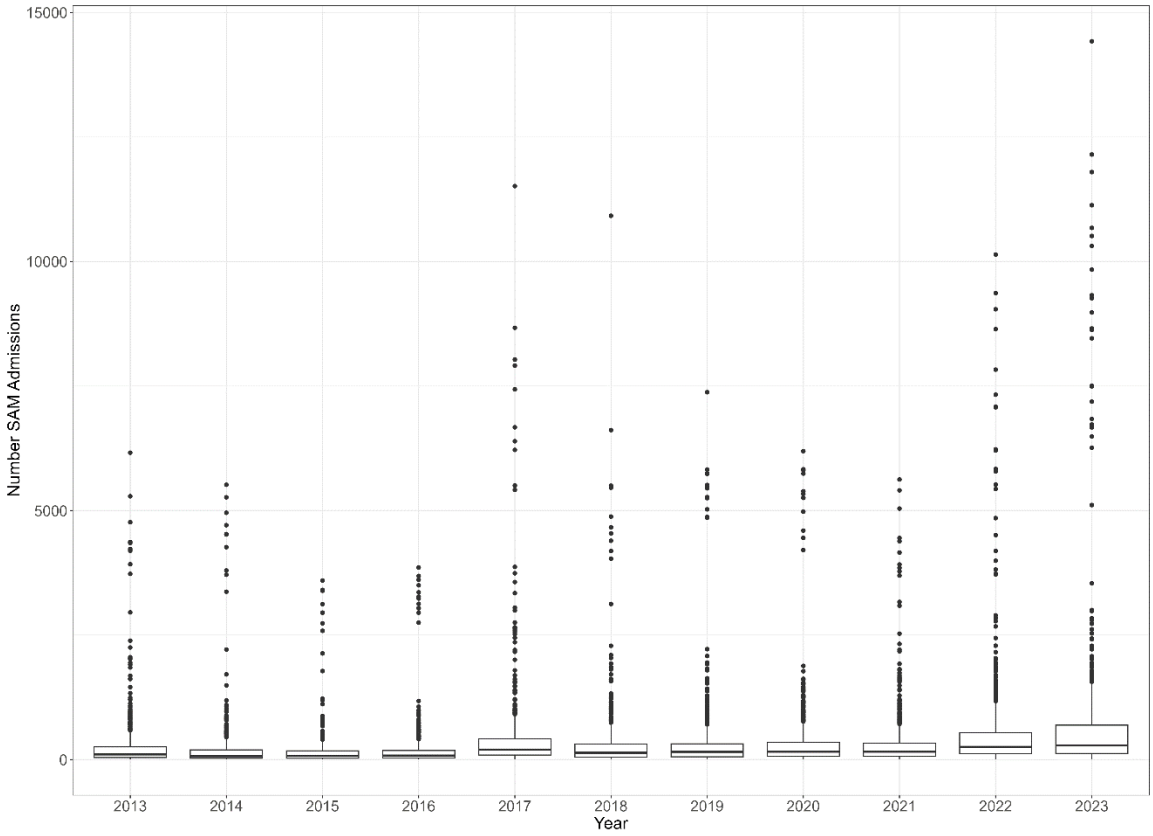


Box plot of the number of Aid Workers Killed per year. Each point corresponds to a district.



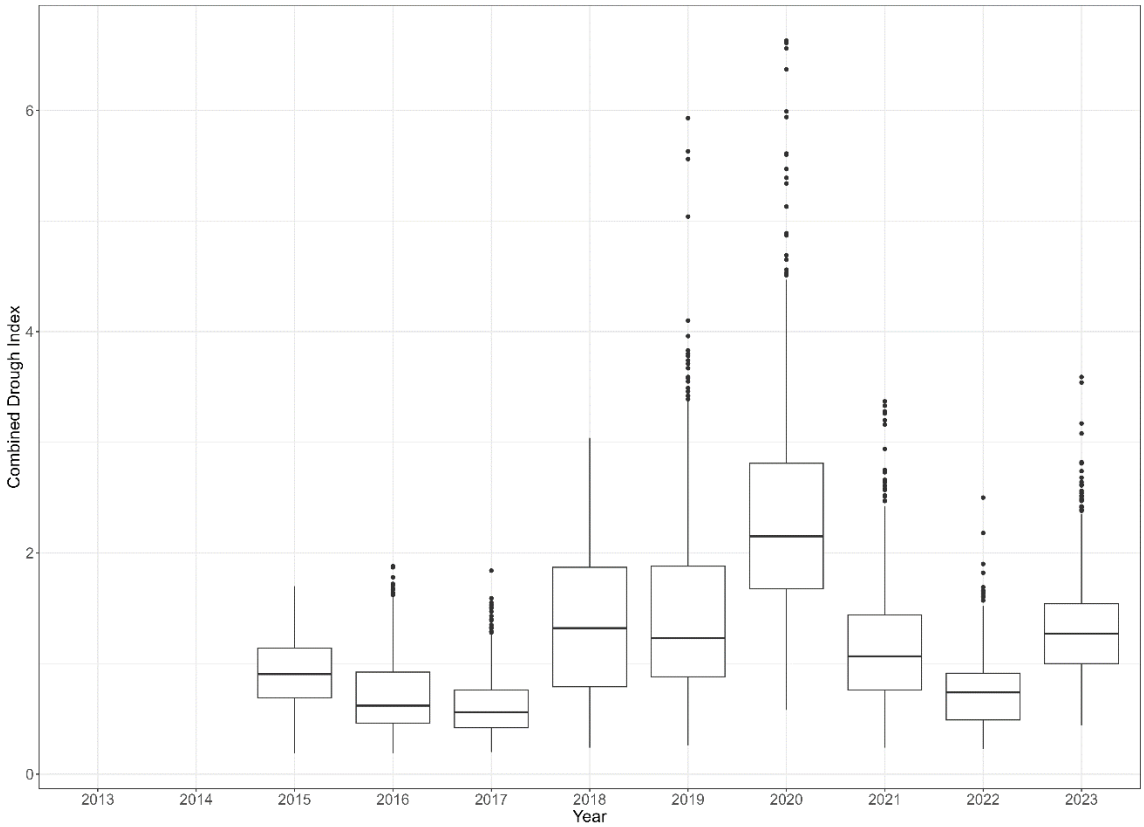
SAM

Box plot of the SAM Admission cases per year. Each point corresponds to a district.



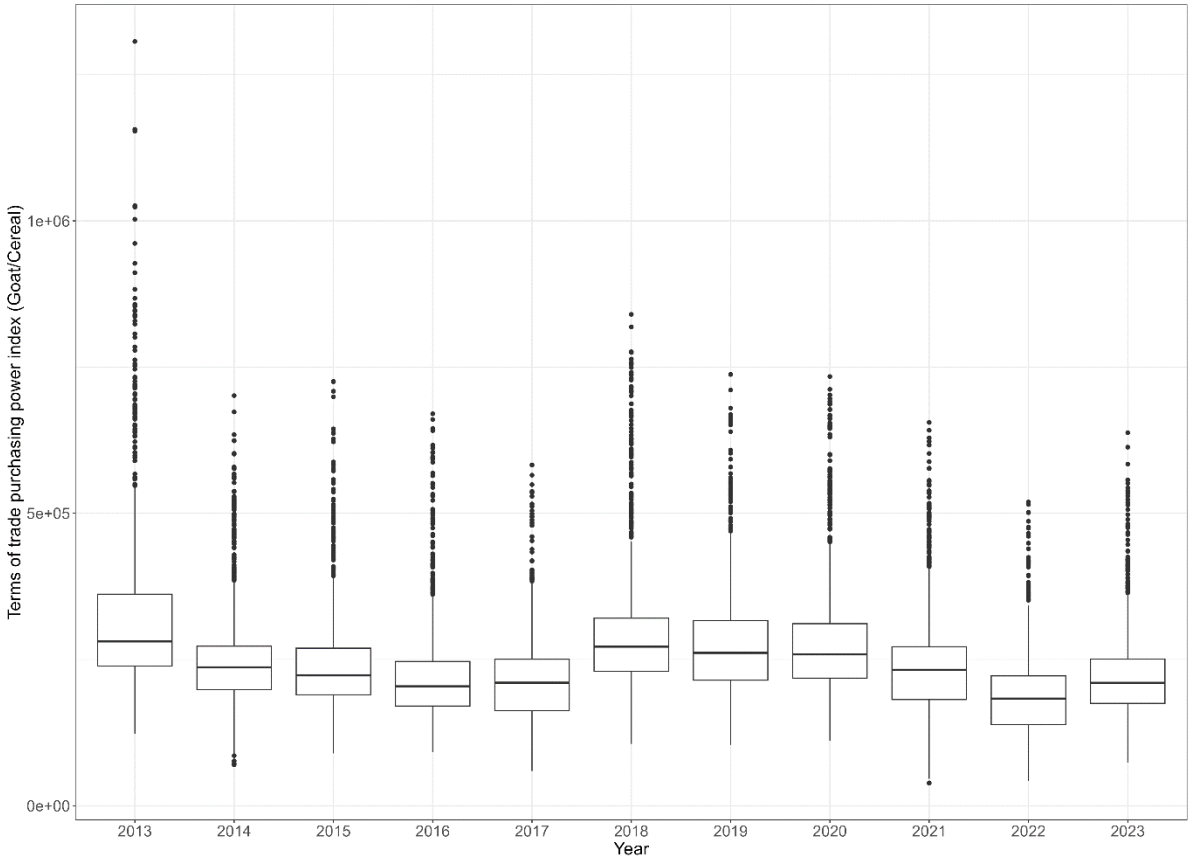
CDI

Box plot of the Combined Drough Index per year. Each point corresponds to a district.

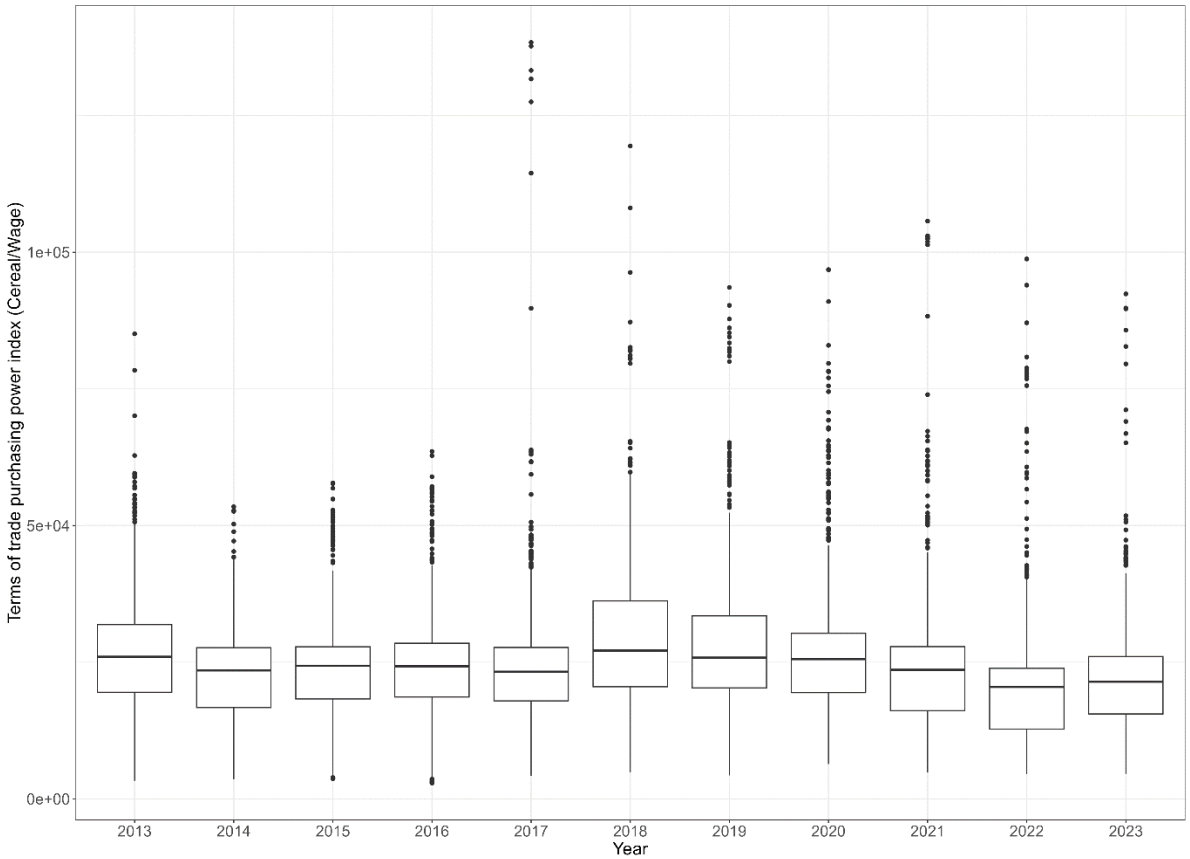


Trade/Power Index

Box plot of the Terms of trade purchasing power index (Goat/Cereal) per year. Each point corresponds

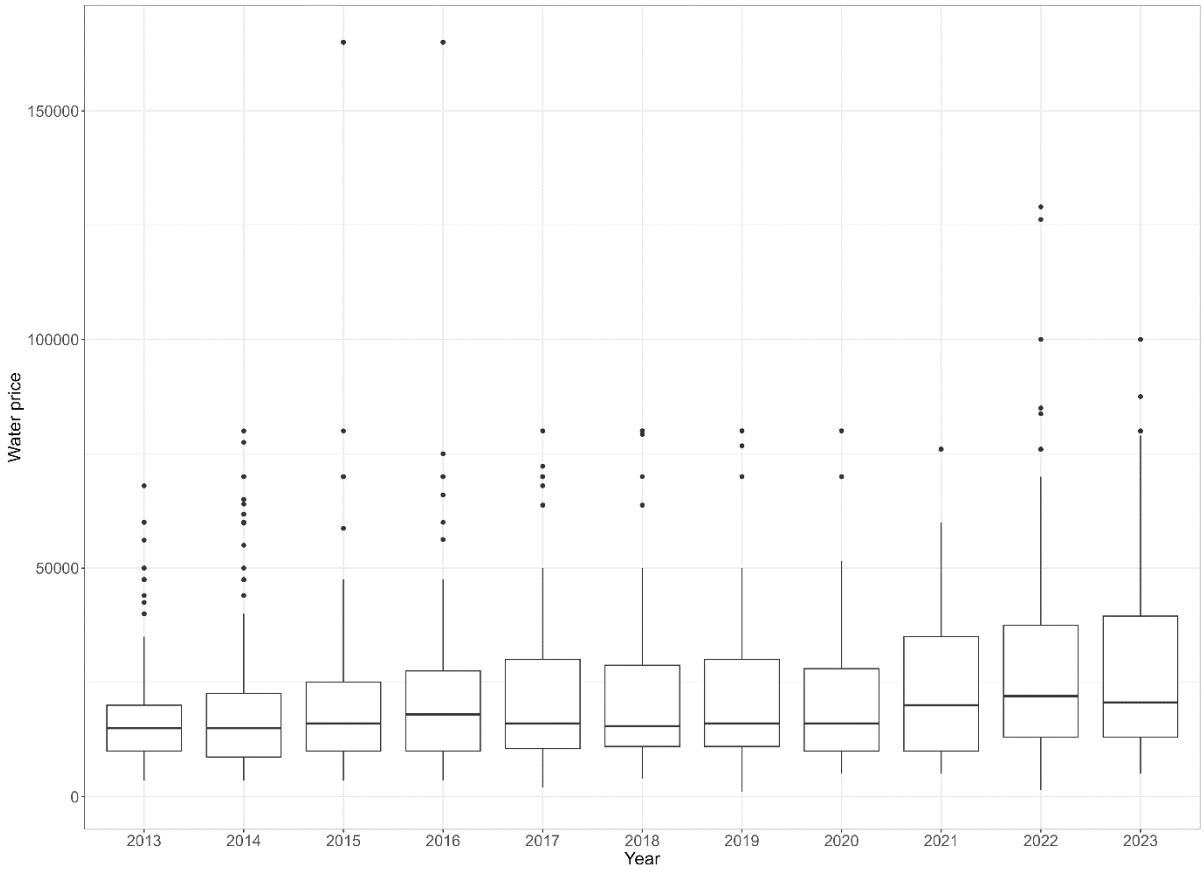


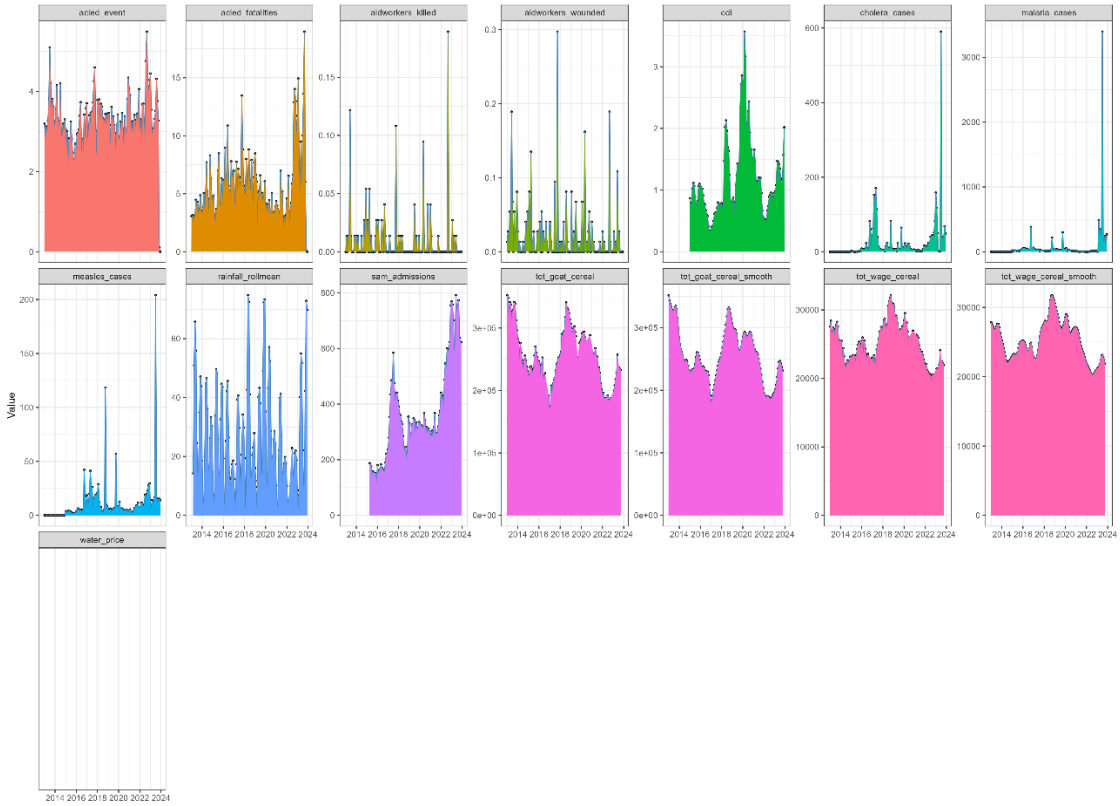
Box plot of the Terms of trade purchasing power index (Cereal/Wage) per year. Each point correspond



Water Price

Box plot of the Water Price per year. Each point corresponds to a district.



All Predictors Distribution

Visualisations of each predictor raw per time unit